

Section 1: Analysis of Indicators of Compromise (IoCs)

The following Indicators of Compromise were identified during the behavioral analysis of the `clientarm4n.elf` malware sample:

- **IoC 1: File Hash (MD5)**
 - **Value:** `4806a550a504c1991566ebe2987b800e`.
 - **Detection Method:** Static File Analysis and Hash Comparison. This hash was compared against global threat databases to determine its malicious status.
 - **Threat Indication:** The unique MD5 hash provides a persistent fingerprint for this malware. Its presence on a system, regardless of the filename used, indicates a confirmed infection by this specific threat actor's toolkit.
 - **IoC 2: Malicious Process Behavior (Network-based)**
 - **Value:**
`https://github.com/Zaeem20/FREE_PROXIES_LIST/raw/refs/heads/master/http.txt`.
+1
 - **Detection Method:** Dynamic Analysis / Process Tree Observation. The process `clientarm4n.elf` (PID 6231) was observed spawning a shell to execute `curl` and `wget` commands targeting this specific URL.
+1
 - **Threat Indication:** This behavior indicates an automated stage of the attack where the malware is attempting to download a proxy list. This is typically used for anonymizing Command & Control (C2) communication or masking further network scanning activities.
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Section 2: MITRE ATT&CK Mapping

The following tactics and techniques observed in the sandbox report have been mapped to the **MITRE ATT&CK Framework**:

- **Tactic: Defense Evasion**
 - **Technique:** Indicator Removal: File Deletion (**T1070.004**).
 - **Observation:** The malware utilized the `rm -f` command to delete temporary files from the `/tmp/` directory, specifically targeting logs and shell scripts created during its initial execution.
- **Tactic: Execution**
 - **Technique:** Command and Scripting Interpreter: Unix Shell (**T1059.004**).
 - **Observation:** The malware repeatedly forked shell processes (`/bin/sh`) to execute complex command strings for system discovery and network fetching.
+2

- **Tactic: Discovery**
 - **Technique:** System Information Discovery (**T1082**).
 - **Observation:** The process was observed executing `df -h` and `uname -p` commands to gather technical specifications about the host system's architecture and storage.
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Section 3: OpenCTI Platform Implementation

This section documents the implementation of the **OpenCTI** Threat Intelligence Platform for the centralization of analyzed data.

- **Platform Setup:** The platform was deployed using a **Docker-Compose** configuration. This includes the core OpenCTI API, a frontend interface, and supporting infrastructure: **Redis** (for state management), **ElasticSearch**(for real-time indexing), and **Minio** (for artifact storage).
- **Connector Integration:** To enable intelligence ingestion, the following connectors were integrated:
 1. **MITRE ATT&CK Connector:** Used to import the latest TTPs, allowing the technical behaviors seen in the `clientarm4n.elf` report to be automatically linked to broader adversarial patterns.
 2. **AlienVault OTX Connector:** Configured to pull real-time "pulses" containing malicious hashes and URLs, ensuring the platform can cross-reference local lab findings with global threat data.
- **Functional Evidence:** The successful setup is evidenced by the "Data Management" dashboard showing active heartbeats from all containers and a populated "Knowledge Graph" where the **MD5 hash** identified in Section 1 is mapped to the **Defense Evasion** techniques identified in Section 2.